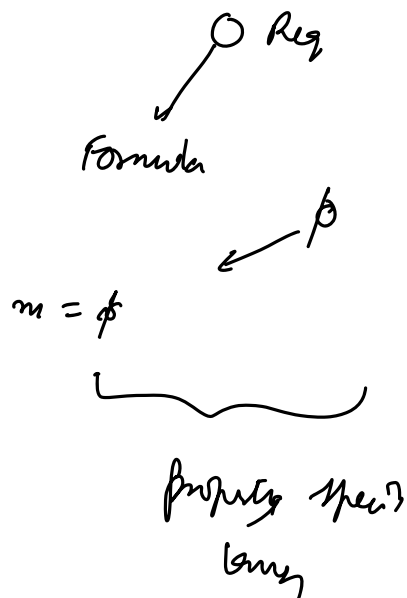
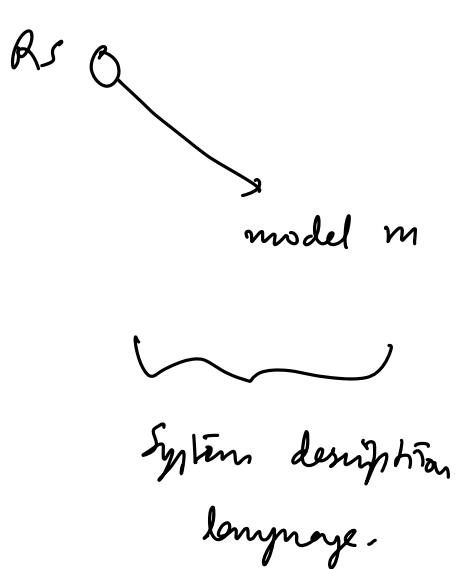


Feb 3, 11:30

CS648 / class 5 / CTL - Syntax - Semantics.pdf

Property specification verification.



System

semantics

PSL

CTL syntax in BNF

$\phi ::=$ ^① true $\mid$ ^② f $\mid$ ^③ $\neg p$
 $\downarrow$
negation

World	
$\mathcal{N}$	
x_1	x_2
Proposition:	
$x_1 > x_2$	

④

$\phi_1 \wedge \phi_2$ conjunction - logical AND

⑤

$\phi_1 \vee \phi_2$ disjunction - logical OR

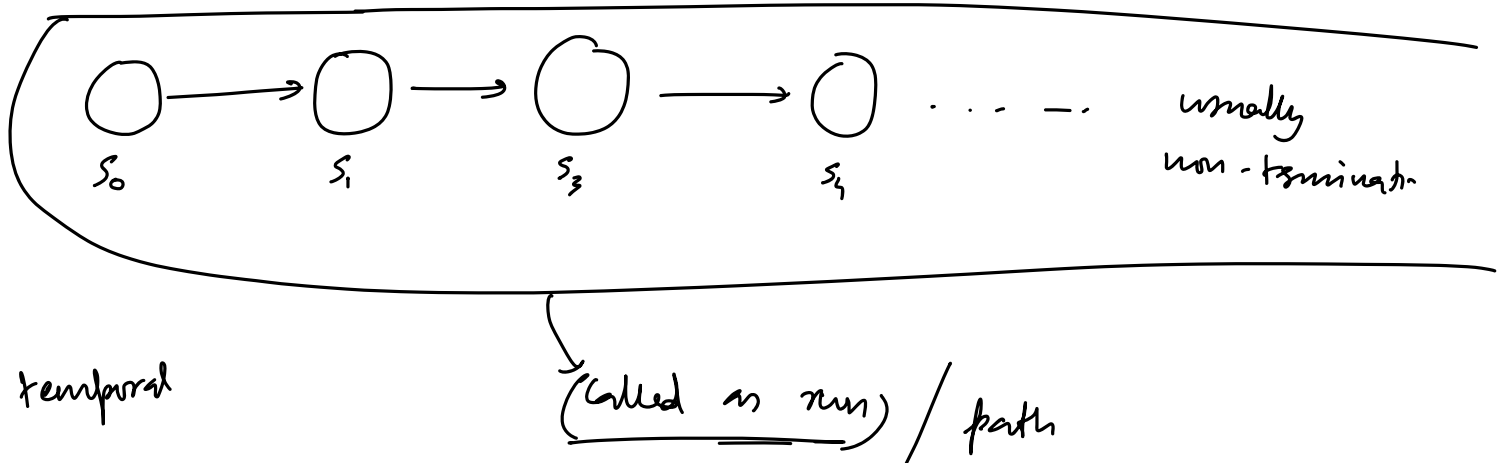
⑥

$\phi_1 \rightarrow \phi_2$ implies

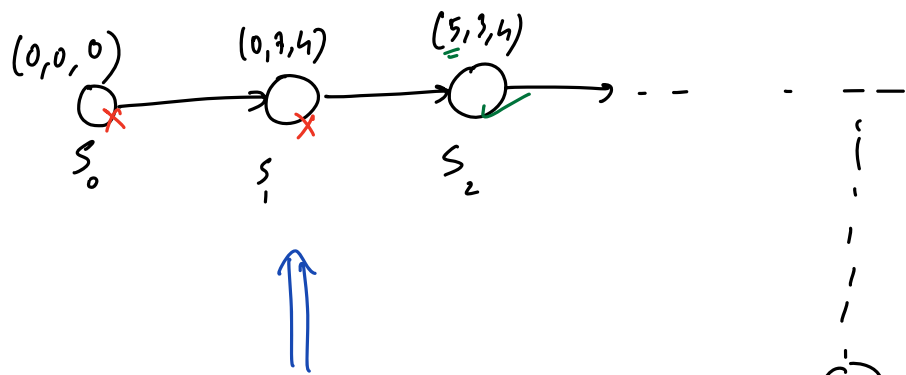
⑦

$\phi_1 \leftrightarrow \phi_2$ double implication

formal language: syntax vs semantics



$AX (x \geq 5)$
 CTL operation predicate



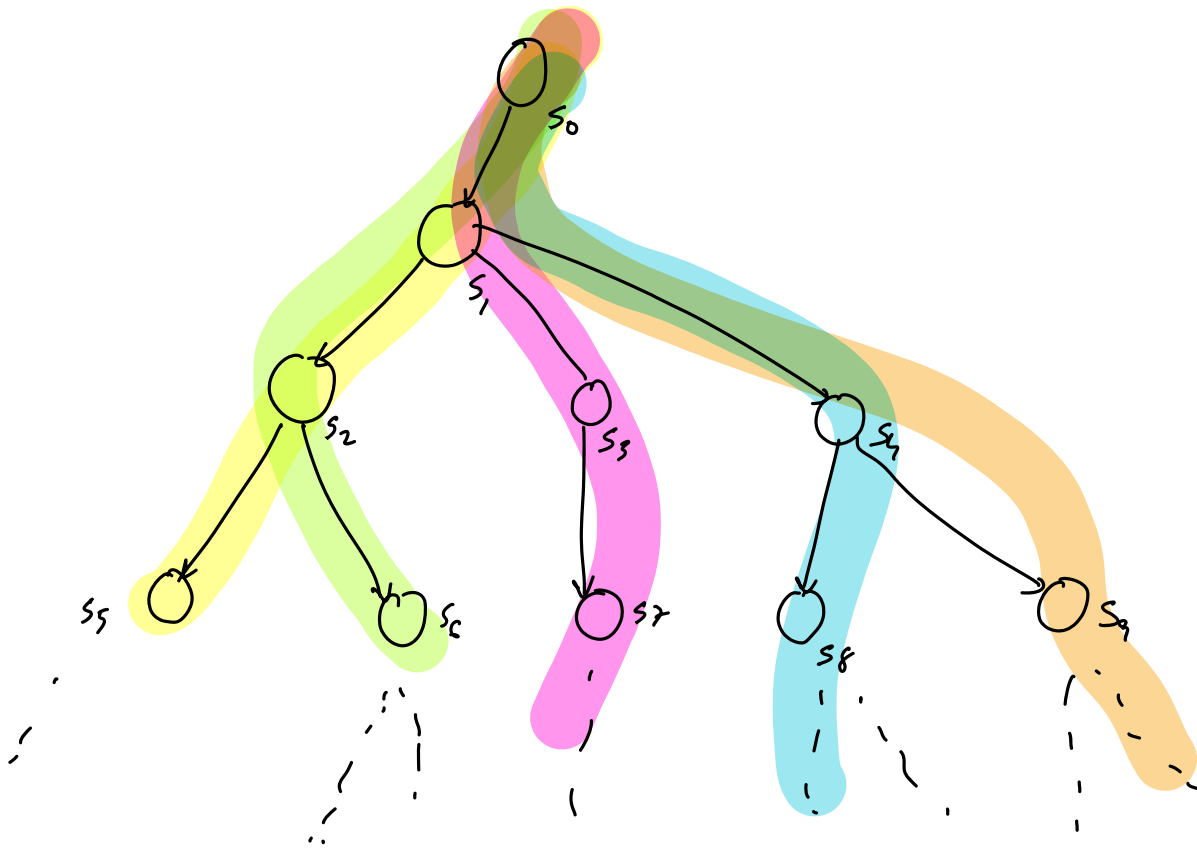
where this formula evaluates to true

x, y, z
 $s_i \dots s_n \dots$

A X
 path state
 quantifying qualifying

$A G$	$E G$
$A F$	$E F$
$A X$	$E X$
$A U$	$E U$
$A R$	$E R$

CTL : Computation Tree Logic



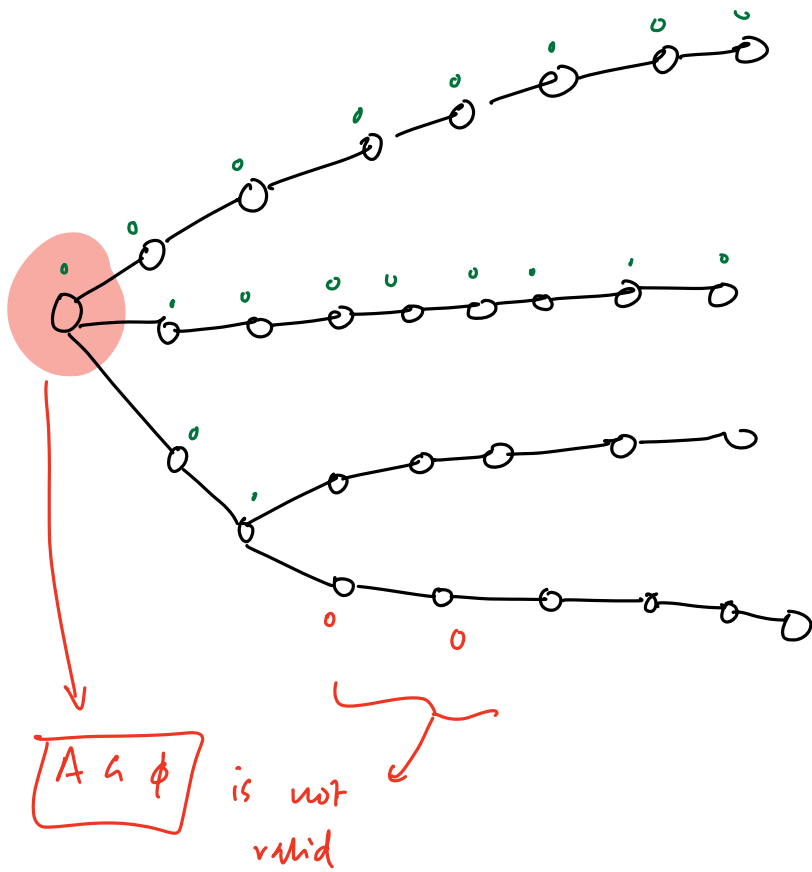
There are many paths rooted in the common root.

a CTL formula

$$AX(\phi)$$



→ among all paths, there is a successor state where ϕ holds.

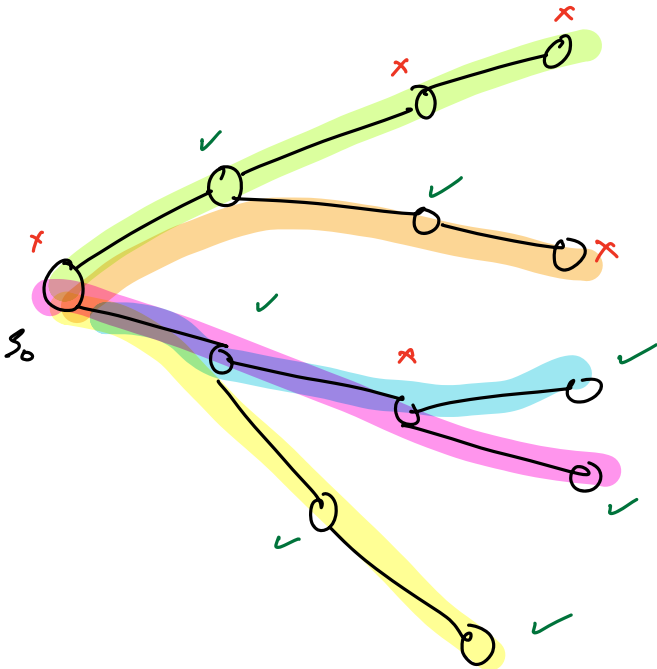


$A \ G \ \phi$

A	: always
G	: global
F	: Future
X	: next
U	: until
R	: Release

$A \ F \ \phi$

$\therefore$ there is a tick on every path,
 $\therefore$ this formula is true at the root

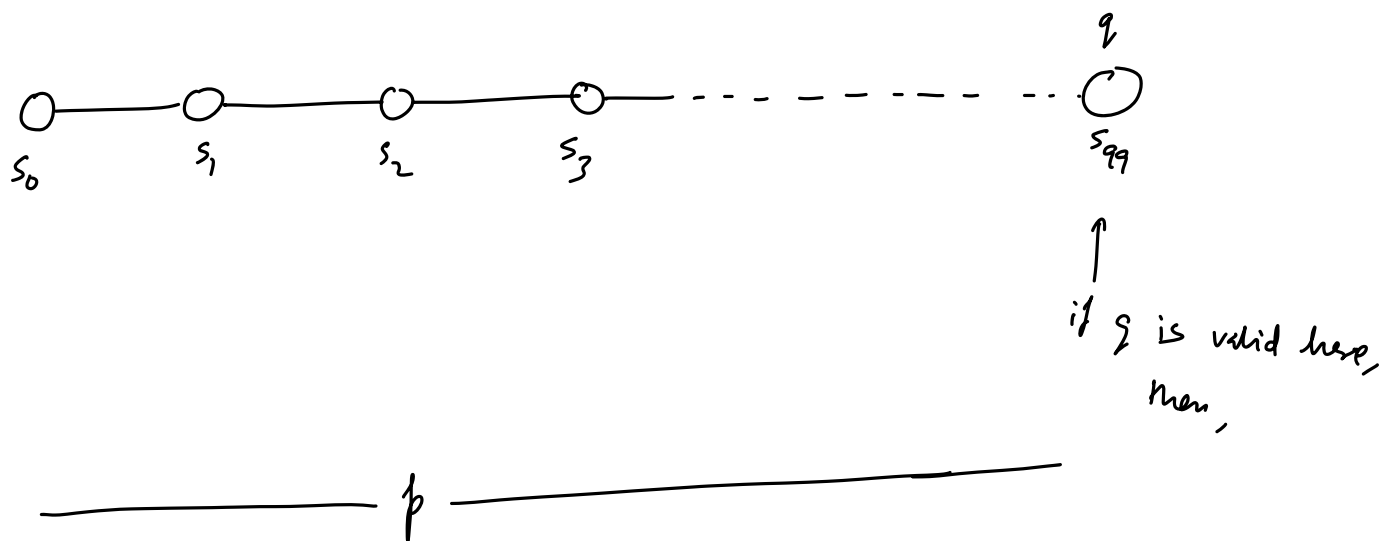


$$A [p \cup q] : A \cup [p, q]$$

↙
"until"

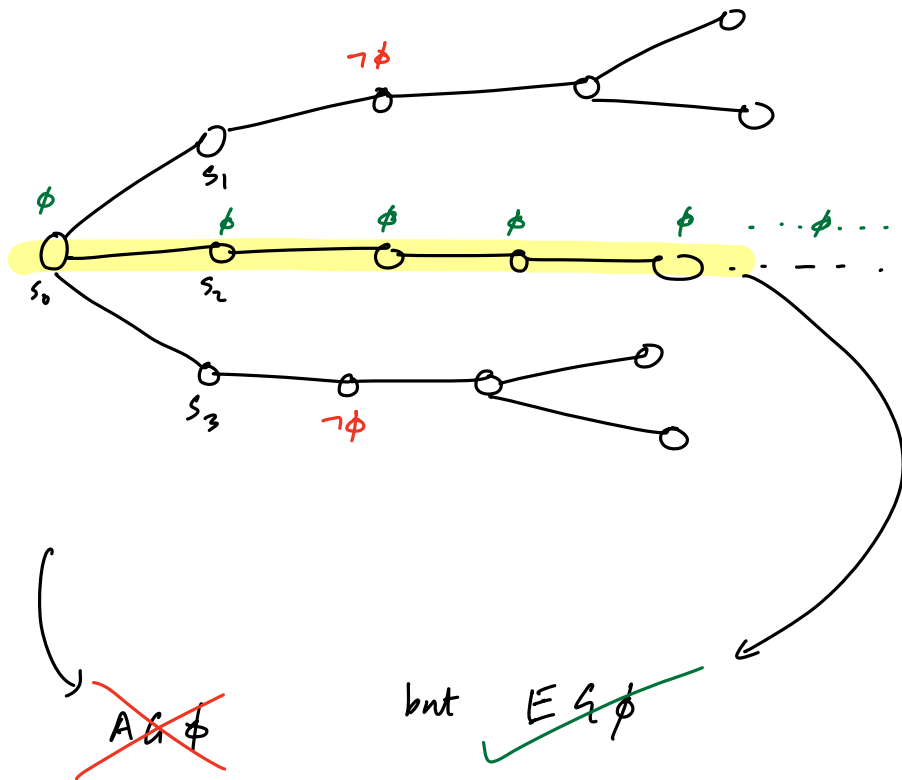
$$A \cup [p, q]$$

$$A [p \cup q]$$



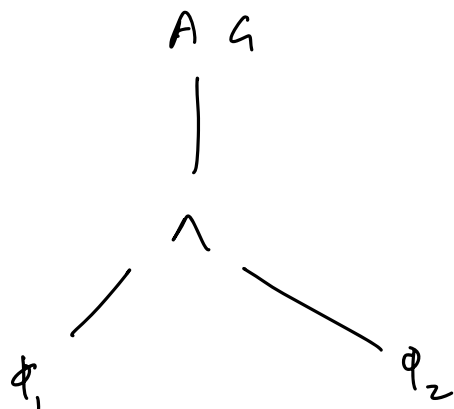
in all the preceding states, p should be valid

computation tree:



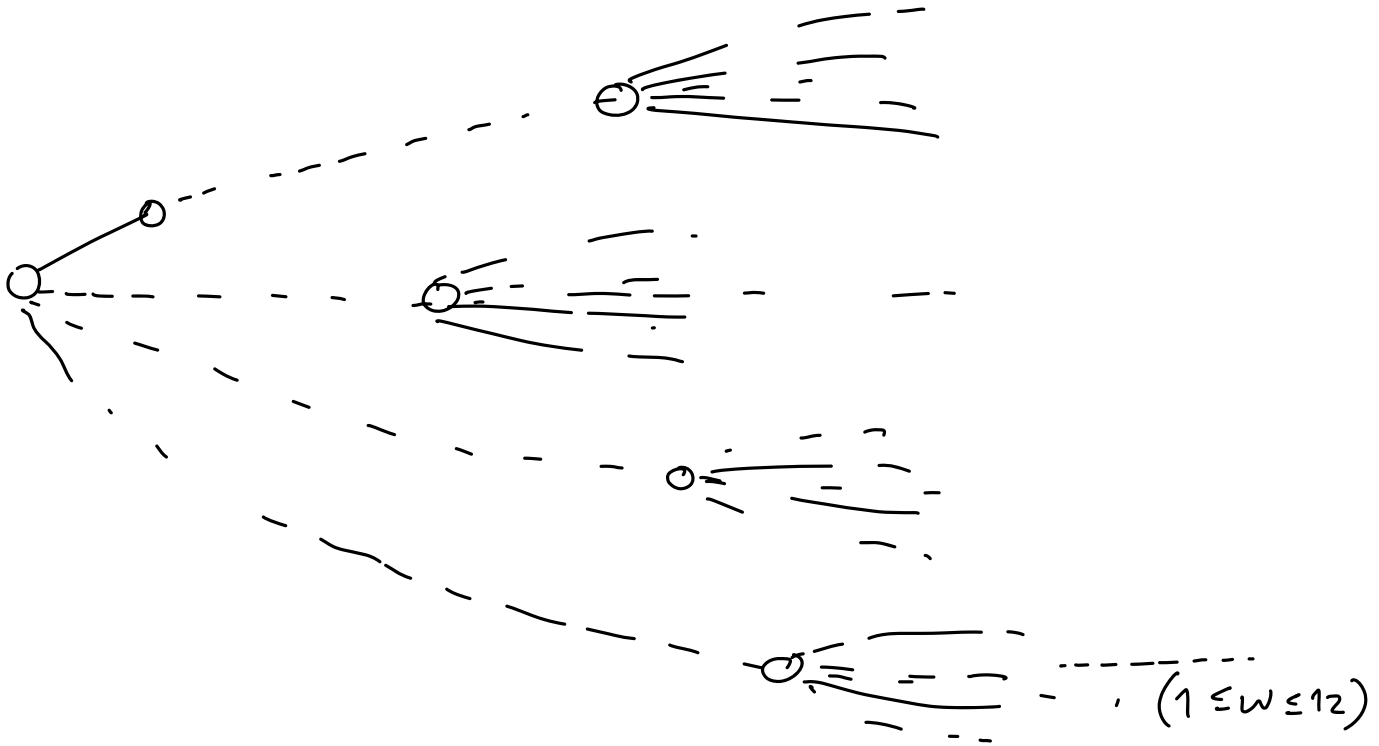
Potential error 1:

Check if formula is well formed or not



$$AG \left(\overbrace{0 \leq w}^{\phi_1} \wedge \overbrace{w \leq 12}^{\phi_2} \right)$$

$$AF \left(AG \left(w \geq 1 \wedge w \leq 12 \right) \right)$$



! or ? $\rightarrow$ here is an intersection array automata

$I_2!$ $\rightarrow$ emitting

? $\rightarrow$ consuming.

Neurachy

Property Specification Languages (Temporal)

temporal

modal- μ -logic

linear Temporal Logic



Branching Temporal Logic



propositional logic

exam may have q outside
slides, like this $\Uparrow$

distinction of temporal logics $\rightarrow$ reading assignment from
Edmund Clarke book.

$AG \text{ } EF \text{ } p$ is weaker than $AF \text{ } p$

$AF \text{ } AG \text{ } p$ is stronger than $FG \text{ } p$